

2. Why is a virtual memory address called 'virtual'?

Why is a virtual memory address called 'virtual'?

Pick **ONE** option

☐ This memory address does not map to actual memory cell that corresponds to address number.

☐ This memory is the same as physical memory.

Solve question 3

☐ This memory cannot be accessed by applications at run time.

☐ This memory address is always present even after a system is restarted.

Clear Selection

3. Consider the two functions incr and decr shown below. incr(){ decr(){ wait(s
are 5 threads each invoking incr once, and 3 threads each invoking decr once
shared variable X is 10 Suppose there are two implementations of the sema
initialized to 1, 1, 2. S is a counting semaphore initialized to 2. Let V1 and V2 b

4. What is the main difference between "user mode" and "kernel mode"?

What is the main difference between "user mode" and "kernel mode" in an operating system?

Pick **ONE** option

- ☐ User mode allows direct hardware access, kernel mode does not.
- ☐ Kernel mode can access protected memory regions, user mode cannot.
- ☐ Kernel mode applications are given less priority than user mode applications.
- ☐ User mode instructions can switch the CPU to kernel mode, but not vice-versa.

Clear Selection

5. Which data structure is best suited for implementing a priority queue?

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<https://www.hackerrank.com/test/g1lj816k4tp/questions/c3aaf70q0e6>

ALL



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5. Which data structure is best suited for imp

Which data structure is best suited for implementing a priority c

Pick **ONE** option



Array



Slack



Heap



Linked List

Clear Selection

6. In the context of Dynamic Programming,

In the context of Dynamic Programming, what is "memoizatio

<https://www.hackerrank.com/test/g1lj816k4tp/questions/a7dn8s3ffm6>

In the context of Dynamic Programming, what is "memoization"?

Pick **ONE** option

- ☐ Writing an algorithm in a memo to remember it
- ☐ Recomputing solutions to subproblems multiple times for accuracy
- ☐ Storing solutions to subproblems to avoid recomputation
- ☐ Optimizing a problem by breaking it into subproblems and solving iteratively

Clear Selection

7. A certain program must store the names of all the people who finish the finish line as the competitors finish. After the race is over the program must output the names of the competitors who placed 1 to 10 and then 100, 200, 300 and so on, as the competitors finish. What data structure would be the best choice for this program?

<https://www.hackerrank.com/test/g1lj816k4tp/questions/asnrkim1c2c>


```
#include <stdio.h>
int main() { char arr[] = "abcd"; char *p = arr;
printf("%c\n", *p); return 0; }
```

```
#include <stdio.h>
int main() {
    char arr[] = "abcd";
    char *p = arr;
    printf("%c\t", **p);
    printf("%c\t", *p++);
    printf("%c\t", (*p)++);
    printf("%c\n", *p);
    return 0;
}
```

Pick **ONE** option

☐ b b b c

☐ b c c d

☐ b b c c

☐ b c c c

Clear Selection

<https://www.hackerrank.com/test/g1lj816k4tp/questions/3g83a54sl0o>

ALL



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10. What is the latest long-term release of Oracle Database?

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Pick **ONE** option

☐ 23ai

☐ 22ai

☐ 24ai

☐ 19ai

Clear Selection

11. Which of the following is not a property of

Which of the following is not a property of Relational Database

<https://www.hackerrank.com/test/g1lj816k4tp/questions/1qed2r1nbff>

2. Which SQL constraint do we use to set some value to a table when value is null?

Which SQL constraint do we use to set some value to a table when value is null?

Pick **ONE** option

☐ NOT NULL

☐ DEFAULT

☐ CHECK

☐ UNIQUE

Clear Selection

13. Truncate table is classified as:

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www.hackerrank.com/test/g1lj816k4tp/questions/1pdsi1nm44h

1. How will you distribute incoming traffic to a set of Web Servers?

How will you distribute incoming traffic to a set of Web Servers?

Pick **ONE** option

- ☐ By deploying a Firewall before the servers
- ☐ By deploying a router before the servers.
- ☐ By deploying a load balancer before the servers.
- ☐ By deploying a switch before the servers.

Clear Selection

15. In a distributed system, what is CAP theorem?

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In a distributed system, what is CAP theorem?

Pick **ONE** option

- ☐ It specifies the maximum allowable response time for a system
- ☐ It describes the trade-offs between consistency, availability, and partition tolerance
- ☐ It defines the minimum number of servers required for fault tolerance
- ☐ It outlines the process for scaling a system horizontally

Clear Selection

16. What is a Zombie process in Linux?

What is a Zombie process in Linux?

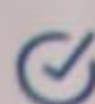
What is a Zombie process in Linux?

What is a Zombie process in Linux?

Pick ONE option

- ☐ A process which is running in the background.
- ☐ A process which is running for a long time.
- ☒ A process whose execution has completed but it still has an entry in the process table.
- ☐ A process which has crashed due to memory corruption.

Clear Selection

 Saved!

17. Which HTTP method is suitable for resource creation?

Which HTTP method is suitable for resource creation?

Pick ONE option

Which HTTP method is suitable for resource creation?

Which HTTP method is suitable for resource creation?

Select **ONE** option

☐ PUT

☐ PATCH

☒ POST

☐ CREATE

Clear Selection

18. Which response status code category represents

29. What is the remainder when 2000^{1000} divided by 13?

What is the remainder when 2000^{1000} divided by 13?

Pick **ONE** option

☐ 3

☐ 4

☐ 8

☐ 11

Clear Selection

30. Interpret Charts

The following graph shows the number of people (in millions) using Hingto the market

<https://www.hackerrank.com/test/g1lj816k4tp/questions/75eqgqdk30>



ALL



1. Rearrange Students

In your school, two lines of students are formed A and B, with each line having exactly N students. The two lines of students will face each other. The Physical Education teacher wants the heights of all students standing across from each other to be equal. To do this, students may be swapped from one line to another. Any student in one line can be swapped with any student in the other line. Each swap counts as one operation, and each operation costs the minimum height in the swap. Reordering the students within one line has no cost. Determine the minimal cost to achieve the desired result. If it is impossible, return -1.

Example

$arrA = [4, 2, 2, 2]$

$arrB = [1, 4, 1, 2]$

If the arrays are sorted descending, they are $arrA = [4, 2, 2, 2]$ and $arrB = [4, 2, 1, 1]$. Swap a 2 in $arrA$ with a 1 in $arrB$ for a cost of $\min(1, 2) = 1$. This is the only swap that needs to occur so the answer is 1.

Function Description

Complete the function `rearrangeStudents` in the editor below.

`rearrangeStudents` has the following parameters:

- `int arrA[n]`, the heights of children in line A
- `int arrB[n]`, the heights of children in line B

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